

Requirements				
Req. ID	Requirement Title	Requirement Intent	Version ID	Status
WMC-REQ-001	System Initialization	On System start, the controller shall enter the Idle stage with all actuators OFF, cycle_complete set to 0, and fault_indicator set to 0.	Ver. 01	
WMC-REQ-002	Start Condition	When start_button is pressed and door_closed is 1, the controller shall start the washing cycle.		
WMC-REQ-003	Door Safety Before Start	If start_button is pressed while door_closed is 0, the controller shall remain in Idle state and not initiate the cycle.		
WMC-REQ-004	Door Lock Activation	After a valid start command, the controller shall activate door_lock and shall wait for door_lock_delay duration before entering the Water_Fill_Wash_Cycle stage.		
WMC-REQ-005	Wash-Cycle Water Fill	During Water_Fill_Wash_Cycle stage, the controller shall turn ON water_valve and shall remain in this stage until water_level_ok becomes 1. The water_level_ok is shared with the rinse-cycle water fill stage.		
WMC-REQ-006	Detergent Fill Stage	After wash-cycle water filling is completed, the controller shall enter Detergent_Fill stage and turn ON detergent_valve until detergent_level_ok becomes 1.		
WMC-REQ-007	Wash Stage	After detergent filling is completed, the controller shall enter the Wash stage and turn on wash_motor for the configured wash_duration.		
WMC-REQ-008	Wash-Cycle Drain	After wash_duration is completed, the controller shall enter Drain_Wash_Cycle stage and turn ON drain_pump for the configured drain_duration. The drain_duration parameter is shared with the Drain_Rinse_Cycle stage.		
WMC-REQ-009	Rinse-Cycle Water Fill	After wash-cycle draining is completed, the controller shall enter Water_Fill_Rinse_Cycle stage and shall turn ON water_valve until water_level_ok become 1. The water_level_ok signal is shared with the wash-cycle water fill stage.		
WMC-REQ-010	Rinse Stage	After rinse-cycle water filling is completed, the controller shall enter the Rinse stage and turn ON wash_motor for the configured rinse_duration.		
WMC-REQ-011	Rinse-Cycle Drain	After rinse_duration is completed, the controller shall enter Drain_Rinse_Cycle stage and turn ON drain_pump for the configured drain_duration. The drain_duration parameter is shared with the Drain_Wash_Cycle stage.		
WMC-REQ-012	Spin Stage	After rinse-cycle draining is completed, the controller shall enter Spin stage and turn ON the spin_motor for the configured spin_duration.		
WMC-REQ-013	Cycle Completion	After spin_duration is completed, the controller shall enter Cycle_Complete stage, turn OFF all actuators, shall unlock the door_lock, and shall set cycle_complete to 1.	Ver. 02	
WMC-REQ-014	Pause Command	If pause_button is pressed during any active cycle, the controller shall enter Paused stage and turn OFF active actuator outputs.		
WMC-REQ-015	Resume Command with History	If resume_button is pressed from the Paused stage and door_closed is 1, the controller shall resume from the previously interrupted cycle stage using Stateflow history junction behavior.		
WMC-REQ-016	Door Safety During Pause	If door_closed is 0 during Paused stage, the controller shall not resume the cycle until door_closed becomes 1, even if resume_button is pressed.		
WMC-REQ-017	Detergent Fill Timeout Fault	If detergent_level_ok does not become 1 within detergent_fill_timeout duration during Detergent_Fill stage, the controller shall enter Fault stage.		
WMC-REQ-018	Generic Fault Detection	If fault_signal becomes 1 during any active cycle stage, the controller shall immediately transition to the Fault stage.		
WMC-REQ-019	Fault Response	In the Fault stage, the controller shall turn OFF water_valve, detergent_valve, drain_pump, wash_motor and spin_motor, and shall set fault_indicator to 1. door_lock shall remain ON in the Fault stage.		
WMC-REQ-020	Fault Reset	If reset_button is pressed in the Fault stage and fault_signal is 0, the controller shall clear fault_indicator and with all outputs reset to Idle stage conditions.		
WMC-REQ-021	Door Unlock at Safe States	The controller shall deactivate door_lock only in the following conditions: when entering Idle stage, when entering Cycle_Complete stage, or when valid fault reset occur (i.e. fault_signal is 0 AND reset_button is pressed in Fault stage).	Ver. 03	
WMC-REQ-022	Idle Output Reset	In the Idle stage, all actuator outputs (water_valve, detergent_valve, drain_pump, wash_motor, spin_motor, door_lock) shall be OFF, cycle_complete shall be 0, and fault_indicator shall be 0.		
WMC-REQ-023	Actuator Mutual Exclusion	The controller shall ensure that wash_motor and spin_motor are never simultaneously ON at any point during operation.		
WMC-REQ-024	Stage Status Output	The controller shall provide stage_status to indicate the currently active washing machine operating stage.		